Assignment 2 Solutions

Problem 1

Given:

The model $(1 - B)(1 - 0.2B)\, X_t = (1 - 0.5B) W_t$

This model can also be written as:

$$X_t - 1.2X_{t-1} + 0.2X_{t-2} = W_t - 0.5W_{t-1}$$

$$\Rightarrow \quad X_t = 1.2X_{t-1} - 0.2X_{t-2} + W_t - 0.5W_{t-1} \qquad - \ (1)$$

(a) The forecasts one- and two-step ahead is given by:

One-Step Ahead:

$$X_{t+1} = 1.2X_t - 0.2X_{t-1} + W_{t+1} - 0.5W_t$$

[By replacing t by $t+1$ in the original model equation (1)]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = 1.2E(X_t|X_1,X_2,\dots,X_t) - 0.2E(X_{t-1}|X_1,X_2,\dots,X_t)$$
$$E(W_{t+1}|X_1,X_2,\dots,X_t) - 0.5E(W_t|X_1,X_2,\dots,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = 1.2X_t - 0.2X_{t-1} - 0.5W_t = 1.2x_t - 0.2x_{t-1} - 0.5w_t$$

[We know that X_t, X_{t-1} and W_t are not independent of $X_1, X_2, \dots, X_t$
and that W_{t+1} is not a function of $X_1, X_2, \dots, X_t$]

Hence, the one step ahead forecast with the given model is $\boldsymbol{\hat{x}_t(1) = 1.2x_t - 0.2x_{t-1} - 0.5w_t}$.

Two-Steps Ahead:

$$X_{t+2} = 1.2X_{t+1} - 0.2X_t + W_{t+2} - 0.5W_{t+1}$$

[By replacing t by $t+2$ in the original model equation (1)]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = 1.2E(X_{t+1}|X_1,X_2,\dots,X_t) - 0.2E(X_t|X_1,X_2,\dots,X_t)$$
$$E(W_{t+2}|X_1,X_2,\dots,X_t) - 0.5E(W_{t+1}|X_1,X_2,\dots,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = 1.2X_{t+1} - 0.2X_t = 1.2\hat{x}_t(1) - 0.2x_t$$

[We know that X_{t+1} and X_t are not independent of $X_1, X_2, \dots, X_t$
and that W_{t+2} and W_{t+1} are not a function of $X_1, X_2, \dots, X_t$]

Hence, the two steps ahead forecast with the given model is $\boldsymbol{\hat{x}_t(2) = 1.2\hat{x}_t(1) - 0.2x_t}$.

To show that the recursive expression for forecasting three or more steps ahead, we can find the three steps ahead forecast and try to see a patter in the forecasting terms.

Three-Steps Ahead:

$$X_{t+3} = 1.2X_{t+2} - 0.2X_{t+1} + W_{t+3} - 0.5W_{t+2}$$

[By replacing t by $t+3$ in the original model equation (1)]

$$\Rightarrow \quad \hat{X}_{t+3} = \hat{x}_t(3) = 1.2E(X_{t+2}|X_1,X_2,....,X_t) - 0.2E(X_{t+1}|X_1,X_2,....,X_t)$$
$$E(W_{t+3}|X_1,X_2,....,X_t) - 0.5E(W_{t+2}|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \hat{X}_{t+3} = \hat{x}_t(3) = 1.2X_{t+2} - 0.2X_{t+1} = 1.2\hat{x}_t(2) - 0.2\hat{x}_t(1)$$

[We know that X_{t+2} and X_{t+1} are not independent of $X_1, X_2, \ldots, X_t$ and that W_{t+3} and W_{t+2} are not a function of $X_1, X_2, \ldots, X_t$]

As we see, the white noise terms start becoming zero, just as in the two-step ahead forecast and the pattern seen is that the forecast for three and more steps ahead depends on the previous two forecasts. That is,

$$\Rightarrow \quad \hat{X}_{t+l} = \hat{x}_t(l) = 1.2\hat{x}_t(l - 1) - 0.2\hat{x}_t(l - 2)$$

This is exactly what we needed to show.
Hence, the forecast for three or more steps ahead can be found using the formula

$$\boldsymbol{\hat{x}_t(l) = 1.2\hat{x}_t(l - 1) - 0.2\hat{x}_t(l - 2)}$$

(b) The variance of the one-, two- and three-steps ahead forecast errors can be found by using the equation (9.3.27) from the textbook $\sigma_w^2 \sum_{j=0}^{l-1} \psi_j^2$

For this particular model, the ψ weights required to find the forecast errors asked for, are given by:

$$\psi_0 = 1$$
$$\psi_1 = \theta_1 + \psi_0\phi_1 = -0.5 + 1.2(1) = 0.7$$
$$\psi_2 = \theta_2 + \psi_1\phi_1 + \psi_0\phi_2 = 0.7(1.2) + (1)(-0.2) = 0.64$$

Hence, the variance of the forecast errors are as follows:

One-Step Ahead

Forecast error: $X_{t+1} - \hat{X}_{t+1} \quad \Rightarrow \quad \text{Var}(X_{t+1} - \hat{X}_{t+1}) = \sigma_w^2 \sum_{j=0}^{0} \psi_j^2 = \sigma_w^2$

Two-Steps Ahead

Forecast error: $X_{t+2} - \hat{X}_{t+2} \quad \Rightarrow \quad \text{Var}(X_{t+2} - \hat{X}_{t+2}) = \sigma_w^2 \sum_{j=0}^{1} \psi_j^2 = \sigma_w^2\left(1 + 0.7^2\right) = 1.49\sigma_w^2$

Three-Steps Ahead

Forecast error: $X_{t+3} - \hat{X}_{t+3}$

$\Rightarrow \quad \text{Var}(X_{t+3} - \hat{X}_{t+3}) = \sigma_w^2 \sum_{j=0}^{2} \psi_j^2 = \sigma_w^2\left(1 + 0.7^2 + 0.64^2\right) = 1.9\sigma_w^2$

(c) Given the values $w_t = 1$, $x_t = 4$, $x_{t-1} = 3$ and $\sigma_w^2 = 2$, in order to find $\hat{x}_t(2)$, we will use the equation we found previously, in part (a).

We know that $\hat{x}_t(2) = 1.2\hat{x}_t(1) - 0.2x_t$ and $\hat{x}_t(1) = 1.2x_t - 0.2x_{t-1} - 0.5w_t$.

Then, subbing in the values given:

$\Rightarrow$ $\hat{x}_t(1) = 1.2x_t - 0.2x_{t-1} - 0.5w_t = 1.2(4) - 0.2(3) - 0.5(1) = 3.7$

and

$\Rightarrow$ $\hat{x}_t(2) = 1.2\hat{x}_t(1) - 0.2x_t = 1.2(3.7) - 0.2(4) = 3.64$

Also, the standard error of the corresponding forecast error is:

$$\sqrt{Var(X_{t+2} - \widehat{X}_{t+2})} = \sqrt{1.49\sigma_w^2} = \sqrt{1.49(2)} = 1.72$$

Therefore, as we needed to show, the two-steps ahead forecast for the given values is 3.64 and the standard error of the corresponding forecast error is 1.72.

Problem 2

Given:

The ARIMA(0,1,1) model $(1 - B)X_t = (1 - \theta B)\, W_t$

This model can also be written as:

$$X_t - X_{t-1} = W_t - \theta W_{t-1}$$

$$\Rightarrow \quad X_t = X_{t-1} + W_t - \theta W_{t-1} \qquad - \ (1)$$

(a) To show that the one-step ahead forecast error is given by $\hat{x}_t(1) = x_t - \theta w_t$, we take the following steps:

$$X_{t+1} = X_t + W_{t+1} - \theta W_t$$

[By replacing t by $t+1$ in the original model equation (1)]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = E(X_t|X_1,X_2,....,X_t) + E(W_{t+1}|X_1,X_2,....,X_t) - \theta E(W_t|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = X_t - \theta W_t = x_t - \theta w_t$$

[We know that X_t and W_t are not independent of $X_1, X_2, ..., X_t$ and that W_{t+1} is not a function of $X_1, X_2, ..., X_t$]

For further step forecasts, we can find two-steps ahead forecast and try to find a pattern for the recursive formula.

The two-steps ahead forecast is:

$$X_{t+2} = X_{t+1} + W_{t+2} - \theta W_{t+1}$$

[By replacing t by $t+2$ in the original model equation (1)]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = E(X_{t+1}|X_1,X_2,....,X_t) + E(W_{t+2}|X_1,X_2,....,X_t) - \theta E(W_{t+1}|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = X_{t+1} = \hat{x}_t(1)$$

[We know that X_{t+1} is not independent of $X_1, X_2, ..., X_t$ and that W_{t+2} and W_{t+1} are not a function of $X_1, X_2, ..., X_t$]

From the above computation we see that the white noise term begins to vanish (equal to 0) as the further terms are not functions of the historical values of X anymore.

Therefore, the forecasts for two and more steps ahead is given by:

$$\boldsymbol{\widehat{X}_{t+l} = \hat{x}_t(l) = \hat{x}_t(l-1)} \text{ for } l \geq 2$$

(b) For this particular ARIMA(0,1,1) or IMA(1,1) model, which is non-stationary, we can find the π-weights as:

$$\pi_0 = -1$$
$$\pi_1 = \phi_1 + \pi_0\theta_1 = -1 + (\theta)(1) = 1 - \theta$$
$$\pi_2 = \phi_2 + \pi_1\theta_1 = (1 - \theta)(\theta) = \theta(1 - \theta)$$

And in general,

$$\pi_j = \theta\pi_{j-1} = \theta^{j-1}(1 - \theta) \text{ for } j \geq 1$$

Then, we can write out the one-step ahead model forecast as an AR(∞) process:

$$\hat{x}_t(1) = (1 - \theta)x_t + (1 - \theta)\theta x_{t-1} + (1 - \theta)\theta^2 x_{t-2} + \ldots$$

This can also be written as:

$$\hat{x}_t(1) = (1 - \theta)x_t + \theta\hat{x}_{t-1}(1)$$

We see that the π-weights decrease exponentially in this case, implying that $\hat{x}_t(1)$ is actually an exponentially weighted moving average.
This is equivalent to exponential smoothing because as we saw, we are assigning exponentially decreasing weights over time to the historical values. This is evident from the fact that the π-weights sum to 1 over infinity:

$$\sum_{j=1}^{\infty} \pi_j = (1 - \theta)\sum_{j=1}^{\infty} \theta^{j-1} = \frac{1 - \theta}{1 - \theta} = 1$$

Therefore, we can say that $\boldsymbol{\hat{x}_t(1) = (1 - \theta)x_t + \theta\hat{x}_{t-1}(1)}$ is equivalent to exponential smoothing.

(c) The ψ-weights of the process are given by:

$$\psi_0 = 1$$
$$\psi_1 = \theta_1 + \psi_0\phi_1 = 1 - \theta$$
$$\psi_2 = \theta_2 + \psi_1\phi_1 + \psi_0\phi_2 = 0 + (1 - \theta)(1) + 0 = 1 - \theta$$
$$\psi_3 = \theta_3 + \psi_2\phi_1 + \psi_1\phi_2 = 0 + (1 - \theta)(1) + 0 = 1 - \theta$$

And in general,

$$\psi_j = 1 - \theta \text{ for } j \geq 1$$

Then, the variance of the l-steps-ahead prediction error is given by the formula $\sigma_w^2 \sum_{j=0}^{l-1} \psi_j^2$.

$\Rightarrow$ $$\text{Var}(X_{t+l} - \widehat{X}_{t+l}) = \sigma_w^2 \sum_{j=0}^{l-1} \psi_j^2$$

$\Rightarrow$ $$= \sigma_w^2 + \sigma_w^2 \sum_{j=1}^{l-1} \psi_j^2$$

[Separating the summation into $j = 0$ and summation from $j = 1$ to l-1]

$\Rightarrow$ $$= \sigma_w^2 + \sigma_w^2 \sum_{j=1}^{l-1} (1 - \theta)^2$$

[Subbing in the ψ-weights we found previously]

$\Rightarrow$ $$\text{Var}(X_{t+1} - \widehat{X}_{t+1}) = \sigma_w^2 + \sigma_w^2(1 - l)(1 - \theta)^2$$

[Taking the $(1 - \theta)^2$ out of the summation and summing 1 from $j = 1$ to l - 1]

Hence, proved.

Problem 3

Given: AR(2) process

$$X_t + \frac{1}{a(a+1)} X_{t-1} - \frac{1}{a(a+1)} X_{t-2} = W_t$$

(a) The AR polynomial of this process is $1 + \frac{z}{a(a+1)} - \frac{z^2}{a(a+1)}$ whose roots can be found the following way:

$$1 + \frac{z}{a(a+1)} - \frac{z^2}{a(a+1)} = 0$$

$$\Rightarrow \quad \frac{z^2}{a(a+1)} - \frac{z}{a(a+1)} - 1 = 0$$

$$\Rightarrow \quad \frac{z^2 - z - a(a+1)}{a(a+1)} = 0$$

From this, $a \neq 0, 1$ as the fraction would then be undefined. And,

$$\Rightarrow \quad z^2 - z - a(a+1) = 0$$

$$\Rightarrow \quad z = \frac{-1 \pm \sqrt{1 + 4a(a+1)}}{2}$$

$$\Rightarrow \quad z = \frac{-1 \pm \sqrt{1 + 4a^2 + 4a}}{2}$$

Which implies that the process is causal if $|z| > 1$. This implies either $-1 + \sqrt{1 + 4a^2 + 4a} > 2$ or $1 + \sqrt{1 + 4a^2 + 4a} > 2$.

First, $1 + \sqrt{1 + 4a^2 + 4a} > 2 \Rightarrow \sqrt{1 + 4a^2 + 4a} > 1 \Rightarrow 1 + 4a^2 + 4a > 1$
$\Rightarrow 4a^2 + 4a > 0 \Rightarrow a > 0$ or $a < -1$

And, $-1 + \sqrt{1 + 4a^2 + 4a} > 2 \Rightarrow \sqrt{1 + 4a^2 + 4a} > 3 \Rightarrow 1 + 4a^2 + 4a > 9$
$\Rightarrow 4a^2 + 4a > 8 \Rightarrow a^2 + a - 2 > 0$
$\Rightarrow a > 1$ or $a < -2$

Combining both the ranges for a, we get that the process is causal if
$a < -2$ and $a > 1$

(b) For the autocorrelation, we first need to find the autocovariance for the process at lag k.

We know that the mean of the process is $E[X_t] = E[-\frac{1}{a(a+1)} X_{t-1} + \frac{1}{a(a+1)} X_{t-2} + W_t]$

$$\Rightarrow \quad E[X_t] = -\frac{1}{a(a+1)} E[X_{t-1}] + \frac{1}{a(a+1)} E[X_{t-2}] + E[W_t]$$

$$\Rightarrow \quad E[X_t] = \mu = 0/(1 + \frac{1}{a(a+1)} - \frac{1}{a(a+1)}) = 0$$

[Making the assumption for weak stationarity for the $AR(2)$ process, we know that. $E[X_t] = E[X_{t-1}] = E[X_{t-2}] = \mu$ where μ is a constant]

Then, the autocovariance function can be found as follows:
Multiply X_t by X_{t-k} and take the expected value of both sides,

$$E[X_t X_{t-k}] = E[-\frac{1}{a(a+1)}X_{t-1}X_{t-k}+ \frac{1}{a(a+1)}X_{t-2}X_{t-k} + W_tX_{t-k}]$$

$$= -\frac{1}{a(a+1)}E[X_{t-1}X_{t-k}]+\frac{1}{a(a+1)}E[X_{t-2}X_{t-h}]+E[W_tX_{t-h}]$$

[As the white noise process is i.i.d and $E[X_t] = 0$, $E[W_tX_{t-h}] = 0$]

$\Rightarrow \quad E[X_t X_{t-k}] = -\frac{1}{a(a+1)}E[X_{t-1}X_{t-k}]+ \frac{1}{a(a+1)}E[X_{t-2}X_{t-k}]$. Then, $Cov[X_t, X_{t-k}] = E[(X_t)(X_{t-k})] = \gamma_k$

$\Rightarrow \quad \gamma_k = -\frac{1}{a(a+1)}\gamma_{k-1} + \frac{1}{a(a+1)}\gamma_{k-2}$ for $k = 1, 2, 3….;$

Then, the autocorrelation is given by dividing the equation by γ_0,

$\Rightarrow \quad \rho_k = -\frac{1}{a(a+1)}\rho_{k-1} + \frac{1}{a(a+1)}\rho_{k-2}$ for $k = 1, 2, 3….;$

(c) The one-step and two-steps ahead forecast of the process at forecast origin X_t are:

One-Step Ahead:

$$X_{t+1} = -\frac{1}{a(a+1)}X_t + \frac{1}{a(a+1)}X_{t-1} + W_{t+1}$$

[By replacing t by $t+1$ in the original model equation]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = -\frac{1}{a(a+1)}E(X_t|X_1,X_2,....,X_t)+\frac{1}{a(a+1)}E(X_{t-1}|X_1,X_2,....,X_t) + E(W_{t+1}|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = -\frac{1}{a(a+1)}X_t + \frac{1}{a(a+1)}X_{t-1} = -\frac{1}{a(a+1)}x_t + \frac{1}{a(a+1)}x_{t-1}$$

[We know that X_t and X_{t-1} are not independent of $X_1,X_2, …. , X_t$
and that W_{t+1} is not a function of $X_1,X_2, …. , X_t$]

Hence, the one step ahead forecast with the given model is $\hat{x}_t(1) = \frac{1}{a(a+1)}(x_{t-1} - x_t)$

Two-Steps Ahead:

$$X_{t+2} = -\frac{1}{a(a+1)}X_{t+1} + \frac{1}{a(a+1)}X_t + W_{t+2}$$

[By replacing t by $t+2$ in the original model equation]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = -\frac{1}{a(a+1)}E(X_{t+1}|X_1,X_2,....,X_t)+\frac{1}{a(a+1)}E(X_t|X_1,X_2,....,X_t) + E(W_{t+2}|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = -\frac{1}{a(a+1)}X_{t+1} + \frac{1}{a(a+1)}X_t = -\frac{1}{a(a+1)}x_{t+1} + \frac{1}{a(a+1)}x_t$$

[We know that X_{t+1} and X_t are not independent of $X_1,X_2, …. , X_t$
and that W_{t+2} is not a function of $X_1,X_2, …. , X_t$]

Hence, the two-steps ahead forecast with the given model is $\hat{x}_t(1) = \frac{1}{a(a+1)}(x_t - x_{t+1})$.

Problem 4

Given: ARMA(2,1) process $X_t - \alpha X_{t-2} = W_t - \beta W_{t-1}$
where $\alpha, \beta \in R$ and $\alpha, \beta \neq 0, \alpha \neq \beta$

(a) The process X_t is causal and invertible if the AR and MA polynomials have roots outside the unit circle.

Causality: AR polynomial $1 - \alpha z^2 = 0$ has the roots $z = \left|\frac{1}{\sqrt{\alpha}}\right|$ which lie outside the unit circle when $1 > |\sqrt{\alpha}|$

Invertibility: MA polynomial $1 - \beta z = 0$ has the root $z = \frac{1}{\beta}$ which lies outside the unit circle when $1 > |\beta|$

Hence the process is causal when $\mathbf{1 > |\sqrt{\alpha}|}$ and invertible when $\mathbf{1 > |\beta|}$, given that $\alpha, \beta \neq 0, \alpha \neq \beta$

(b) Given the observations $X_1, X_2, ..., X_t$, the best prediction for X_{t+1} and X_{t+2} at the forecast origin X_t can be found in the following way:

Prediction for X_{t+1}:

$$X_{t+1} = \alpha X_{t-1} + W_{t+1} - \beta W_t$$

[By replacing t by $t+1$ in the original model equation]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = \alpha E(X_{t-1}|X_1,X_2,....,X_t) + E(W_{t+1}|X_1,X_2,....,X_t) - \beta E(W_t|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides with the given observations]

$$\Rightarrow \quad \widehat{X}_{t+1} = \hat{x}_t(1) = \alpha X_{t-1} - \beta W_t = \alpha x_{t-1} - \beta w_t$$

[We know that X_{t-1} and W_t are not independent of $X_1, X_2, ..., X_t$
and that W_{t+1} is not a function of $X_1, X_2, ..., X_t$]

Hence, the one step ahead prediction for X_{t+1} with the given model is $\boldsymbol{\hat{x}_t(1) = \alpha x_{t-1} - \beta w_t}$

The **forecast error** for the prediction is given by

$$X_{t+1} - \widehat{X}_{t+1} = \alpha X_{t-1} + W_{t+1} - \beta W_t - \alpha X_{t-1} + \beta W_t = \boldsymbol{W_{t+1}}$$

Prediction for X_{t+2} :

$$X_{t+2} = \alpha X_t + W_{t+2} - \beta W_{t+1}$$

[By replacing t by $t+2$ in the original model equation]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = \alpha E(X_t|X_1,X_2,....,X_t) + E(W_{t+2}|X_1,X_2,....,X_t) - \beta E(W_{t+1}|X_1,X_2,....,X_t)$$

[Taking conditional expectation of both sides with the given observations]

$$\Rightarrow \quad \widehat{X}_{t+2} = \hat{x}_t(2) = \alpha X_t = \alpha x_t$$

[We know that X_t is not independent of $X_1, X_2, ..., X_t$
and that W_{t+2} and W_{t+1} are not a function of $X_1, X_2, ..., X_t$]

Hence, the one step ahead prediction for X_{t+2} with the given model is $\boldsymbol{\hat{x}_t(2) = \alpha x_t}$

The **forecast error** for the prediction is given by

$$X_{t+2} - \widehat{X}_{t+2} = \alpha X_t + W_{t+2} - \beta W_{t+1} - \alpha X_t = \boldsymbol{W_{t+2} - \beta W_{t+1}}$$